

MODY AI

02JSKOV - Human Computer Interaction (2025/2026)

FINAL REPORT

1. Project and Team	1
2. Problem/solution overview	2
3. Needfinding	3
a. Domain	3
b. Interviews	3
c. Synthesis	9
d. Solutions	11
4. Tasks and Storyboard	13
a. Tasks	13
b. Storyboard	13
5. Low-fidelity Prototypes	15
a. Modalities	15
b. Paper prototypes	16
c. Heuristic Evaluations	22
d. Selection	26
6. Medium to High-Fidelity Prototype	27
a. Tools Used	27
b. Screens Selected and Violations Fixed	27
c. Figma Link	28
d. Violation Handling Plan	28
7. High-fidelity Prototype	30
a. Tech Stack	30
b. Github Repo	30
c. Design Choices	30
d. Most Significant Screens	30
e. Standards & Constraints	35
f. Limitations	35
8. Usability testing	36
a. Preparation and run	36
b. Results	40
9. Conclusions	45
a. Learnings	45
b. Feedbacks	45

1. Project and Team

Project name: Draw Companion

Value Proposition: For beginner artists and hobbyists who feel stuck or fear judgment, Draw Companion provides a gentle, reflective companionship that helps them find inspiration and track their own progress to grow with confidence.

Team: Mody AI

Members:

- Zuhal Didem Aytac
- Özgün Koçak
- Mehmed Yusa Erguven
- Mohammadreza Babaei

2. Problem/solution overview

Problem

Based on our interviews, the biggest barrier for beginner painters/drawers is emotional, not just technical. Key Pain Points are

- Fear of Failure: Beginners have a "fear of failure, not being good enough" and are often "too careful".
- Lack of Safe Feedback: They need "honest but encouraging" feedback, not harsh evaluation.
- Feeling Stuck: Beginners often "don't know what to practice next" and need a way to "find inspiration easily".
- Lack of Motivation: They need to see their own "visible progress" to stay motivated.

Solution

We are designing Draw Companion, a "Reflective Drawing Companion" app.

- It's designed to support the emotional side of learning to draw by focusing on personal reflection, AI-driven feedback, and private progress tracking.
- This approach directly addresses the key user needs for safe feedback, inspiration, and motivation that we discovered in our research.

3. Needfinding

a. Domain

Domain: Learning to Draw as a Beginner

Many people are interested in drawing but find it difficult to improve on their own. Without clear guidance or feedback, it can be hard to know what to practice or how to start.

We chose this domain because learning to draw involves a visible output that requires reflection and correction, which makes the process different from other types of self-learning. We wanted to understand how beginners approach drawing, what challenges they face, and how they find motivation or inspiration to continue.

b. Interviews

i. Participants

- P1 - Selen Yukeb - amateur painter who paints as a hobby (immediate user)
- P2 - Ewa Walkovska - art instructor and owner of a new art studio in Nizza, Torino (domain expert)
- P3 - Giuseppe Martano - He improved his skills after series of art classes during middle and high school (lead user)
- P4 - Selin Ozciftci - Exhibiting artist and art instructor in Turkey (domain expert)

ii. Methodology and Execution

Two team members were present at each interview: one leading the conversation, one taking notes.

- Interview 1: Didem conducted the interview, and Yusa took notes.
- Interview 2: Didem conducted the interview, and Yusa connected remotely on mobile to take notes. This included a short observation during an art workshop.
- Interview 3: Özgün conducted the interview, and Mohammadreza took notes.
- Interview 4: Yusa conducted the interview while the rest of the team joined remotely. All participants gave written consent for recording and data use.

Why they were chosen: They represent a diverse set of perspectives: from beginners struggling to start, to an experienced teacher who mentors others, and a passionate self-taught artist. This diversity helped us capture the emotional and practical needs of learners at different stages.

Recruitment: Participants were reached through personal connections and local art communities. One interview included an on-site observation during an art workshop at Ewa's studio.

Consent Forms

[P1 - Consent Form \(PDF\)](#)

[P2 - Consent Form -\(PDF\)](#)

[P3 - Consent Form \(PDF\)](#)

[P4 - Consent Form \(PDF\)](#)

Questions:

Participant 1:

- How did you first become interested in drawing?
- What made you start drawing again after stopping for a while?
- How often do you draw, and when do you usually find time for it?
- What kind of things do you usually like to draw?
- How do you decide what to draw each time
- Can you describe what happens when you sit down to draw — from start to finish?
- What do you find most difficult while drawing?
- How do you know when a drawing is finished?
- Where do you look for guidance or feedback on your drawings?
- Have you ever considered taking a class or joining a drawing group? Why or why not?
- How do you notice improvement in your drawings over time?
- What kind of environment helps you draw best?
- What motivates you to keep drawing even when you're busy or not satisfied with your results?
- Is there something that makes learning or improving more difficult for you?
- How do you feel when you compare your work to what you see online?
- What would make drawing as a hobby more enjoyable or easier to continue long term?
- If you could get help with something in drawing, what would it be?
- Is there anything else you'd like to add that we didn't cover?

Participant 2:

- How did you start teaching drawing, and what led you to open this workshop?
- What kinds of people usually attend your classes?
- What do you think attracts people to learn drawing here instead of online?
- From your experience, what usually motivates people to start learning to draw?
- What are the most common difficulties beginners face in the first lessons?
- How do you notice when someone starts improving or gaining confidence?
- How do you usually structure a beginner course or a typical class?
- Are there specific exercises or activities that work especially well for beginners?
- How do you give feedback to your students during or after class?
- What kind of feedback do students respond to best?
- How do you help students stay motivated when they get frustrated?
- What do you enjoy most about teaching drawing?
- How do you see the role of technology or digital tools in helping people learn art today?
- Is there anything else you'd like to add that we didn't cover?

Participant 3:

- How did you get started with painting?
- What made you interested in painting?
- How long have you been painting, and how often do you do it?
- What do you enjoy most about painting?
- What are the hardest parts or most frustrating moments?
- How do you usually learn or improve your painting skills? (e.g., YouTube, classes, self-practice)
- What kinds of exercises or projects do you do to learn?
- What's something you wish was easier to learn?
- How do you know if you're improving?
- What materials or tools do you use (brushes, paints, tablets, etc.)?
- Do you paint digitally, physically, or both?
- Have you ever tried any digital painting tools, VR or AR apps?
- How does your painting environment affect your experience (lighting, space, mess, etc.)?
- Would you be interested in an AR/VR environment to help you paint or learn? Why or why not?

Participant 4:

- Can you tell me about how you first got into drawing or painting?
- What made you want to continue — what hooked you early on?
- When you were learning, what parts of the process felt the most challenging or discouraging?
- Was there a specific turning point or experience that helped you grow as an artist?
- How has your relationship with drawing or painting changed over the years — from learner to professional?
- What inspired you to start sharing your creative process through workshops?
- What kinds of people usually join your workshops, and what do you think draws them to it?
- From your observations, what usually motivates people to start learning to draw or paint?
- What are some common challenges or fears you see in beginners?
- How do you notice when someone starts improving or gaining confidence?
- How do you usually structure a beginner course or a typical class?
- How do you balance teaching technique and encouraging creativity?
- Are there specific exercises or activities that work especially well for beginners?
- How do you give feedback to your students during or after class?
- What kind of feedback do students respond to best?
- How do you help students stay motivated when they get frustrated?
- What do you enjoy most about teaching drawing?
- In your view, what makes a good drawing teacher?
- How do you see the role of technology or digital tools in helping people learn art today?

- Is there anything else you'd like to add that we didn't cover?

Observation details can be found in: [A1](#)

iii. Results

During the interviews, we found that beginners' main barriers are often emotional rather than purely technical. Participants described fear of failure and judgment, difficulty deciding what to practice next, and a strong need for feedback that is honest but encouraging. Motivation increases when learners can see visible progress and feel supported by a safe community or a trusted mentor. Digital tools can help with inspiration and structure, but participants emphasized that they should support the human and emotional side of learning to draw.

Consent forms used for the interviews are included in the appendix.

iiii. Key Quotes

Participant 1:

"When I sit down to draw, I reconnect with myself and disconnect from everything else."

"Sometimes I feel stuck because I don't know what to practice next."

"I usually search for inspiration on Pinterest when nothing around me inspires me."

"I would love to get help when I can't draw something I imagine, but I don't know who to ask."

Participant 2:

"Many people start classes to get a creative outlet, a space that's different from their everyday analytical work."

"The biggest challenge for beginners is being too careful because they compare themselves to what they see online."

"When students stop worrying about the final result, that's when they really start improving."

"For me, art is everything, my work, my hobby, my way to connect with people."

Participant 3:

"The thing that I struggle most is not being able to represent my imagination on the canvas"

"I prefer calm environments; I can't paint in noisy places."

"Oil painting is hard, one part of the painting can take days to dry."

"I like to paint landscapes, even if I'm not there and see a picture of the place."

"I'd love to learn the techniques that Monet was using in his paintings."

Participant 4:

"I was a very emotional child and in daily life being emotional was sometimes overwhelming. But when I was drawing, this emotion turned into something positive."

"Many people just want to reconnect with themselves." (answer is about motivations of people to go to workshops)

“The biggest fear of beginners is usually the fear of failure, not being good enough.”

“They respond best to honest but encouraging feedback when it's both positive and specific, it reaches to them.”

“I also noticed that public appreciation like hanging their works on the wall for a mini exhibition builds confidence.”

“Patience and empathy are essential. It's not about showing how talented you are but about helping others discover their own potential.”

iiii. Our findings

Emotional connection matters more than skill

Drawing is often a personal and emotional activity rather than a purely technical one. Learners value self-expression, relaxation, and enjoyment over mastery.

(Seen in P1 and P3: drawing for calm and focus; P2 and P4: encourages emotional confidence in students.)

Beginners fear judgment and lack feedback

Many hesitate to draw because they fear failure or criticism. They want guidance and reassurance, not evaluation.

(Seen in P1: avoids public feedback; P2 and P4: needs tools for gentle, quick feedback; P3: His superior classmates intimidated him.)

Tools and materials can be barriers

Uncertainty about how to use tools and limited access to materials often discourage practice. Learners need easier ways to experiment safely.

(Seen in P1: afraid to waste materials; P3: The nature of some elements makes them harder to practice ; P2: bridges traditional and digital media.)

Community and feedback improve learning

Learning with others boosts motivation and helps overcome self-doubt. Regular, kind feedback keeps learners engaged between sessions.

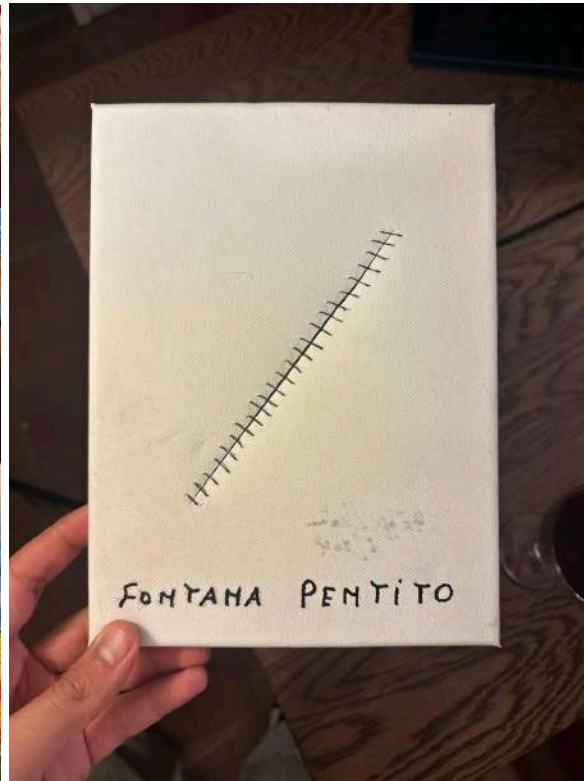
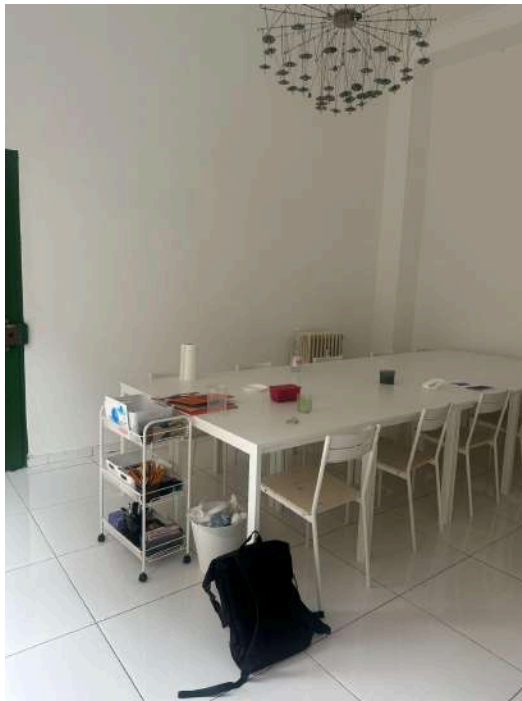
(Seen in all participants: desire for shared spaces or teacher-student connection; P2 stresses motivation between classes.)

Digital tools help, but can't replace the human touch

Technology can support creativity by offering inspiration or structure, but it should never replace the personal or emotional side of art.

(Seen in P1: uses Pinterest for ideas; P2 and P3: value balance between digital aids and traditional practice.)

iiii. Pictures and Artifacts





c. Synthesis

i. Need Extraction

After conducting the interviews, we synthesized the collected data through a brainstorming session. Each of the team members reviewed the interview notes and key quotes individually and identified common patterns and pain points. Afterwards, we discussed these insights together and clustered them into common patterns related to emotions, motivation, feedback, and learning barriers. Based on this process, we translated the main patterns into a set of core user needs that guided the subsequent design decisions.

ii. User Needs

P1

- Needs a way to find inspiration easily when she feels stuck.
- Needs to be able to get feedback and guidance without joining a formal class.
- Needs a way to organize her drawing routine and stay consistent over time.
- Needs to be able to experiment with materials safely without wasting them

P2

- Needs a way to help students overcome fear of failure and perfectionism.
- Needs tools to give quick, personalized feedback in group sessions.
- Needs a way to keep students motivated between classes.
- Needs to be able to bridge traditional art with modern digital tools effectively.

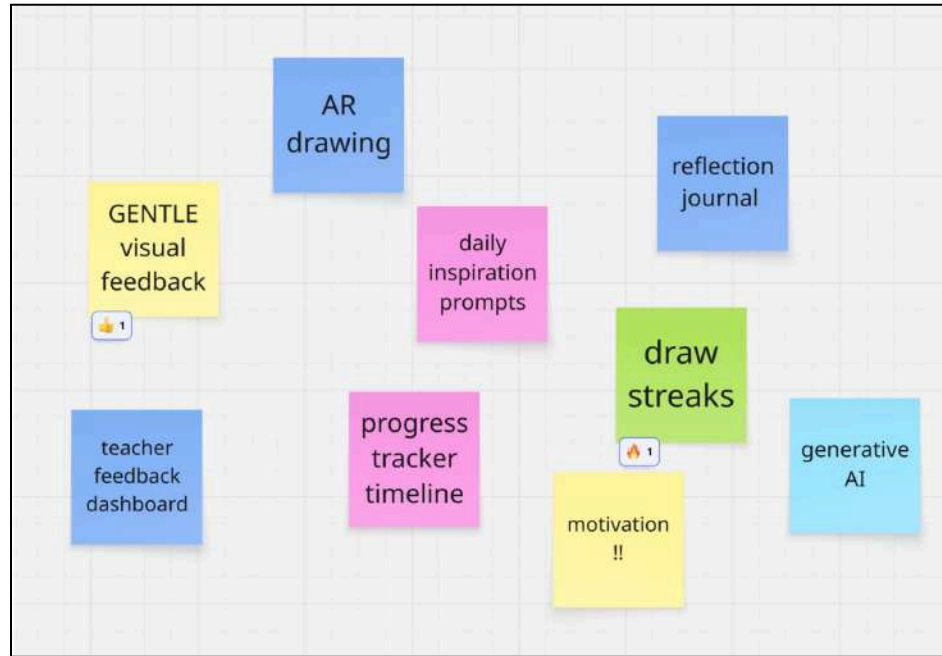
P3

- Needs to learn new techniques (like oil painting) in a simpler, more guided way.
- Needs to practice in calm, focused environments without distractions.

- Needs a way to see his own improvement over time visually.
- Needs to use technology as a helper, but not a replacement for real art.

P4

- Needs a way to keep students motivated between classes.
- Needs personalized ways to overcome the fear of failure



iii. Deep User Needs

Need 1: Needs a way to find inspiration easily when she feels stuck

Need 2: Needs to be able to get feedback and guidance without joining a formal class.

Need 3: Needs a way to stay motivated

Need 4: Needs a way to explore and improve artistic techniques.

These needs were selected because they repeatedly emerged across multiple interviews and directly reflect the main struggles and motivations we observed: beginners often feel stuck and look for inspiration , they want accessible and encouraging guidance without formal classes , they struggle with fear of judgment and need confidence-building support to stay motivated over time , and they want a structured way to improve techniques while keeping the experience calm and enjoyable.

d. Solutions

i. Solutions to Deep User Needs

Need 1: Needs a way to find inspiration easily when she feels stuck

- Encourage artists to share unfinished works in communal spaces so others can continue them.
- Introduce a shared system of rotating daily themes within a group or workshop to generate new ideas and routines.
- Create “inspiration collections” curated by teachers or peers, sets of reference ideas grouped by emotion, style or theme.
- Use generative AI to reinterpret a user’s past work into new styles, helping them see their own work from a fresh perspective
- Introduce an AI “idea partner” that generates visual prompts inspired by the user’s previous drawings, mood, or chosen themes

Need 2: Needs to be able to get feedback and guidance without joining a formal class.

- Encourage online community events, where participants can bring their work, observe others, and discuss feedback casually.
- Develop a way for artists to receive visual suggestions on their work through shared digital or printed annotations.
- Use generative AI to highlight areas of a drawing that differ from reference examples, offering gentle hints rather than grading or critique.
- Use generative AI to visualize multiple possible next steps for a drawing, helping the artist choose their own direction instead of being told what’s wrong.
- An online community where the amateurs can publish their works anonymously and can get feedbacks from professionals.

Need 3: Needs a way to stay motivated

- Encourage artists to look back at earlier drawings in guided reflection sessions to recognize visible progress and personal growth.
- Create small, self-chosen drawing challenges or goals that focus on consistency rather than results.
- Organize recurring community drawing nights where participants share their process and experiences to sustain motivation together.
- Introduce short reflection prompts after each drawing, inviting users to note their emotions, energy, or inspiration level.
- Use generative AI to suggest new creative challenges or techniques based on the user’s past works and current interests.

Need 4: Needs a way to explore and improve artistic techniques.

- Create guided experimentation sessions where artists test new materials or techniques under supervision, reducing waste and frustration.

- Encourage hybrid practice routines that combine digital sketching for planning with traditional painting for final results.
- Use generative AI to simulate outcomes of different materials or brush techniques before applying them on canvas.
- Offer simple proportion and color-planning exercises supported by AR or VR overlays, helping learners visualize structure and tone before starting.
- Use generative AI to suggest small technical exercises tailored to the artist's preferred materials or current skill focus.

We created the proposed solutions through a collaborative brainstorming session together starting from the identified deep user needs and interview outcomes. For each need, the team asked “How might we support this need in a gentle and non-judgmental way?” and explored both social and technological approaches. Ideas were inspired by participants’ quotes, existing drawing practices (for example workshops, peer feedback), and the potential of digital and AI-based tools. The final solutions were selected and refined based on their relevance to the user needs, feasibility, and alignment with the emotional goals of the project.

ii. Selected Solution

Chosen Solution: “DrawCompanion”: A Reflective Drawing Companion

DrawCompanion is a gentle digital companion designed to help beginner and hobbyist artists grow through reflection, inspiration, and community support, not evaluation.

It encourages the emotional and technical sides of drawing, helping users express themselves freely while feeling guided, understood.

Connected User Needs

- Inspiration: Provides personalized prompts and AI reinterpretations of the suggested paintings.
- Feedback: Offers on-demand suggestions and letting users remain in control of their learning process.
- Motivation: Reflective progress tracking and milestones to help artists appreciate growth over time.
- Technique: Digital simulations and guided exercises encourage skill-building

How and Why We Selected It

We selected this solution because every participant expressed the need for feedback that feels safe and supportive rather than evaluative. The idea merges emotional motivation with gentle, on-demand AI support, making it innovative but still feasible to prototype. It's achievable with existing AI image analysis and prompt systems, while keeping the focus on human creativity and emotional comfort. The concept “together” reflects both community and companionship, whether with peers or the AI assistant that learns and grows with the user.

4. Tasks and Storyboard

a. Tasks

Simple Task: Upload a drawing.

Moderate Task: Find inspiration for a new drawing.

Complex Task: Revisit older drawings of the same type to notice progress and stay motivated.

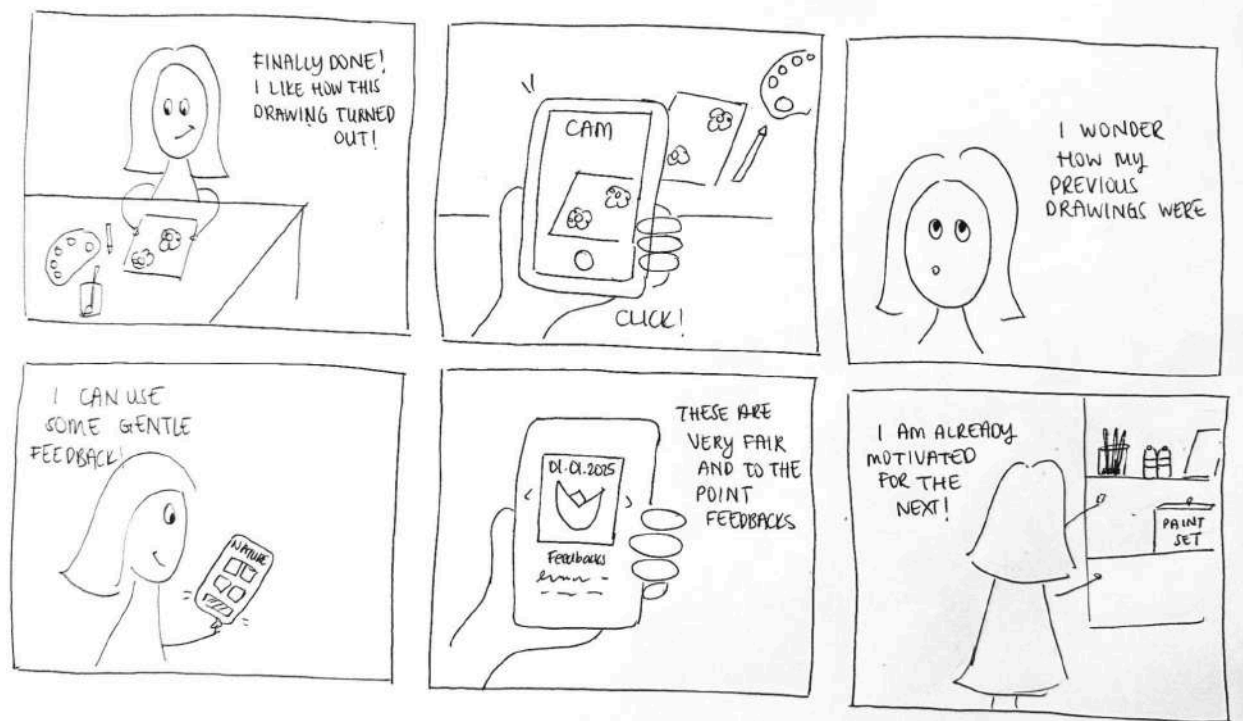
We chose these tasks because they match what beginner and hobbyist artists usually do and what they struggle with.

Simple task (Upload a drawing): We picked this because it's the easiest way to start using the app. Beginners need a quick and low-stress action, and uploading a drawing is a normal first step that helps them enter the app without confusion.

Moderate task (Find inspiration): We chose this because many beginners get stuck and don't know what to draw next. This task is important because it helps them restart and gives them ideas when motivation is low.

Complex task (Revisit older drawings to see progress): We included this because beginners often feel unsure about their skills. Looking at older drawings helps them notice improvement over time, which can increase confidence and keep them motivated.

b. Storyboard



Why did we come up with this solution?

We chose to show the photo upload and progress tasks. Because they follow each other naturally and reflect the main goal of our objective.

Strengths and weaknesses?

It does not cover all three tasks. However, it achieves to show the 2 tasks in a natural and explanatory flow.

Does it achieve the identified goal/user needs?

The current flow touches the core flow of our solution. The covered needs are; Progress Tracking and Need for Feedback.

5. Low-fidelity Prototypes

a. Modalities

We explored several modalities to determine the best fit for our users' needs of privacy, reflection, and ease-of-use.

- Desktop/Web Application
- Smartwatch
- Augmented/Virtual Reality (AR/VR)
- **Mobile Application**
- **Tablet Application**

Mobile Phone (Primary Modality)

We chose the mobile phone for its ease of access at any time. Our Simple Task (Upload a drawing) requires a camera. The mobile phone is the only modality that is always with the user, allowing them to instantly capture their artwork the moment they finish it. This seamless, "in-the-moment" capture is critical for building a consistent reflective progress page.

Target Device Details:

- Form factor: 4-7 inch touchscreen phone
- Platform: IOS/Android
- Main reason: supports fast camera/gallery upload and short, frequent interactions.
- Typical interactions: taking a photo, selecting from camera roll, tapping through short flows, saving entries.
- Usage context: short sessions (1-5 minutes), often right after drawing or when seeking quick motivation/inspiration.

Tablet (Secondary Modality)

Motivation: We selected the tablet because it already has the capabilities that we need in a mobile phone such as a camera which we need for our Simple Task (Upload a drawing) and also its large screen is purpose-built for visual tasks. This can be useful for our Moderate Task (Find inspiration) and Complex Task (Revisit Progress). The larger format allows users to see their artwork in high detail and compare past work side-by-side, which is essential for "noticing progress and staying motivated."

Target Device Details:

- Platform: iPadOS / Android
- Form factor: 8-12 inch touchscreen tablet
- Main reason: larger display supports detailed viewing and progress review.
- Typical interactions: browsing inspiration, viewing drawings in detail, photo upload from gallery/camera supported.
- Usage context: longer sessions (5-15 minutes), typically at home, school, or a study setting.

Future considerations

AR/VR: could be explored later as an optional enhancement for inspiration browsing (Task 2), but it is not a priority for the current scope.

Out of scope

Desktop/Web: not selected because our core usage relies on quick camera capture and mobile-first behavior.

Smartwatch: not suitable due to limited screen size and interaction constraints for image-heavy tasks.

b. Paper prototypes

i. Paper Prototypes

i.i) Mobile Prototype

Home and Related Pages

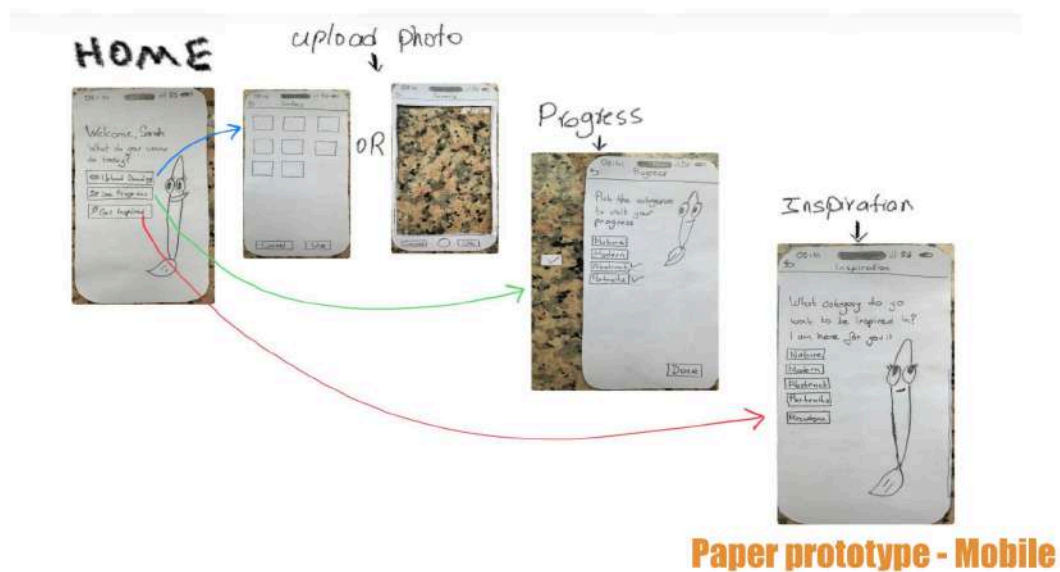
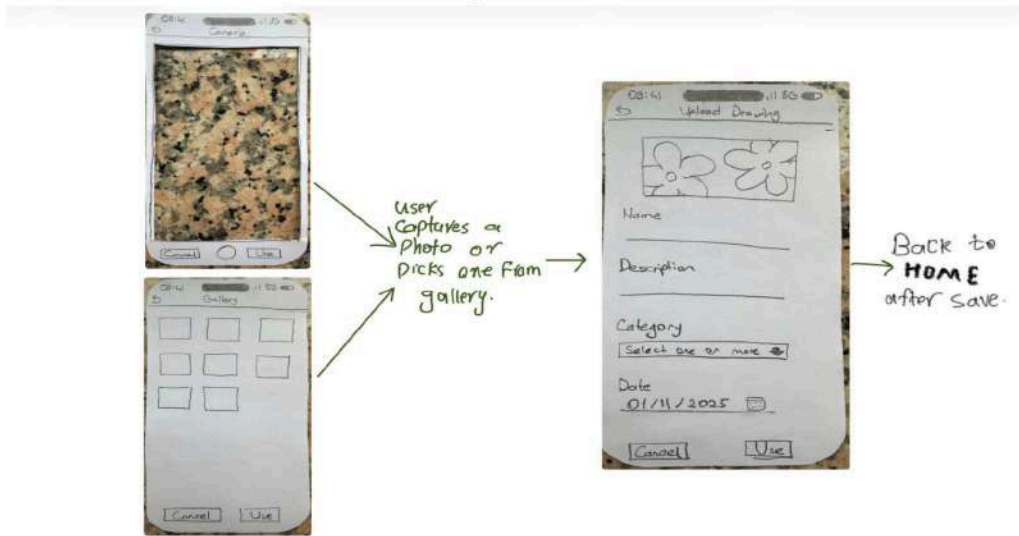
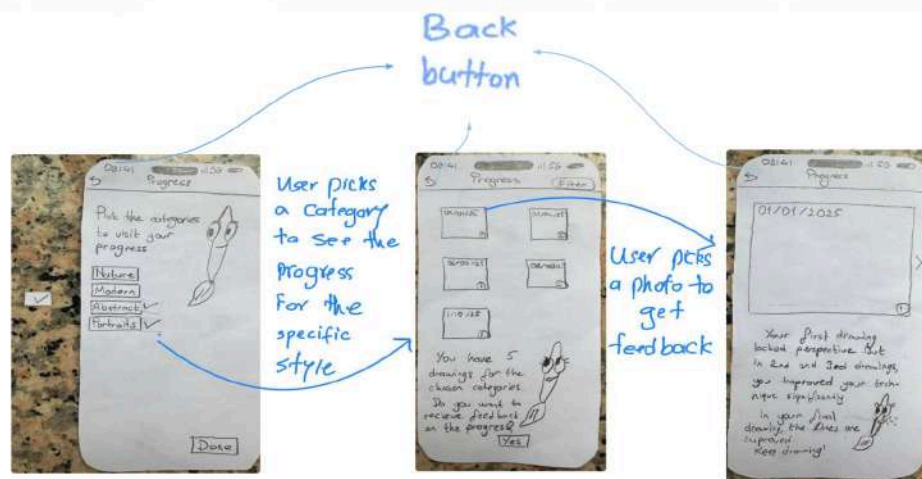


Photo Upload Flow



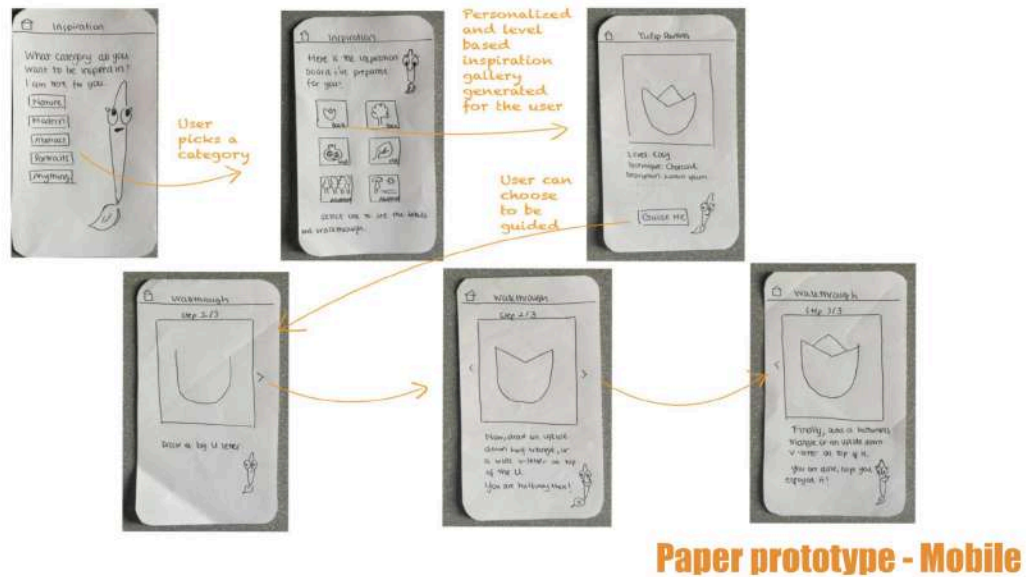
Paper prototype - Mobile

Progress Tracking Flow



Paper prototype - Mobile

Inspiration Flow



I.ii) Tablet Prototype

Home and Related Pages

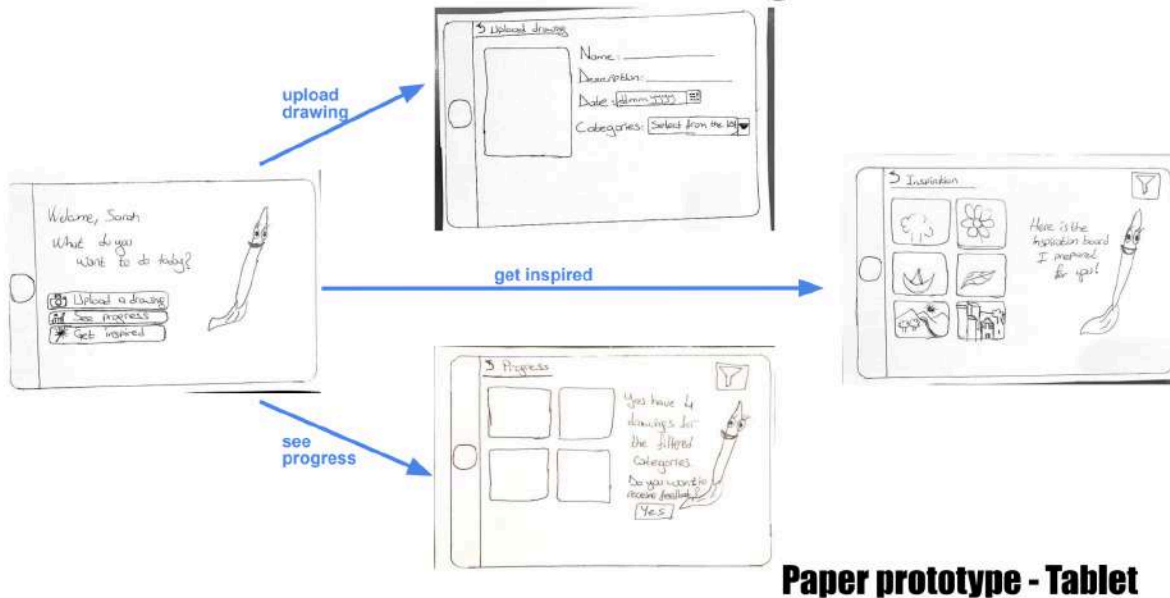
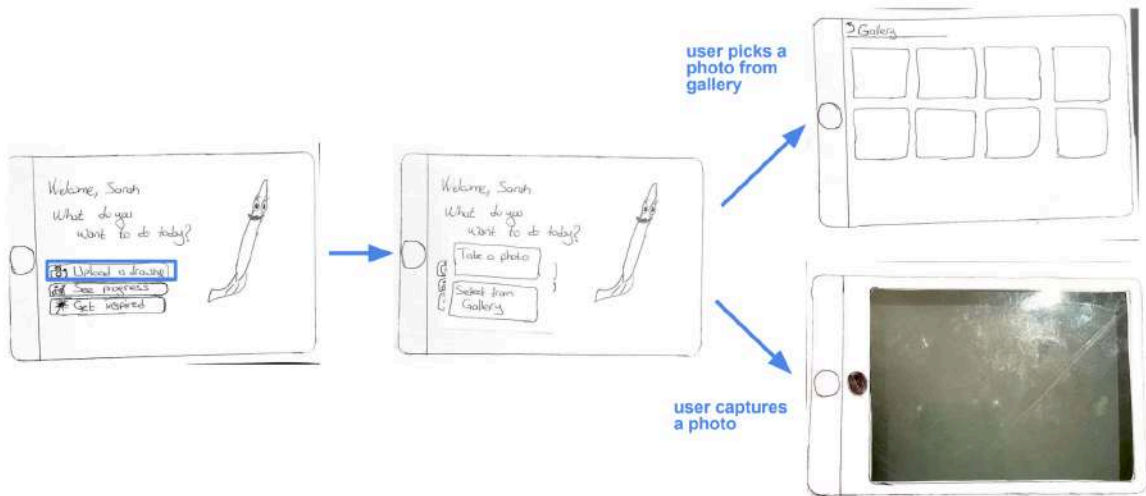
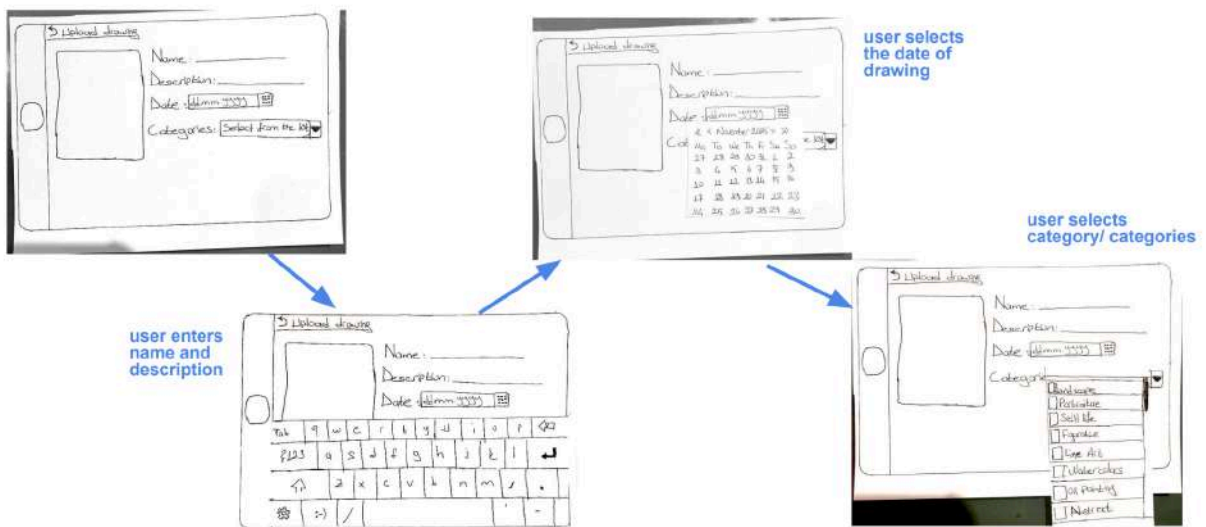


Photo Upload Flow - 1



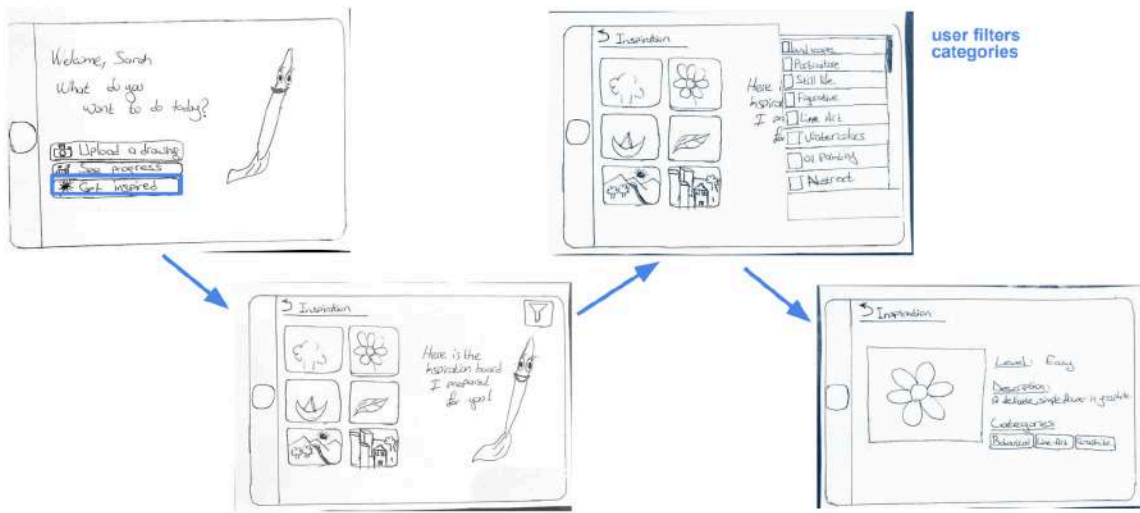
Paper prototype - Tablet

Photo Upload Flow - 2



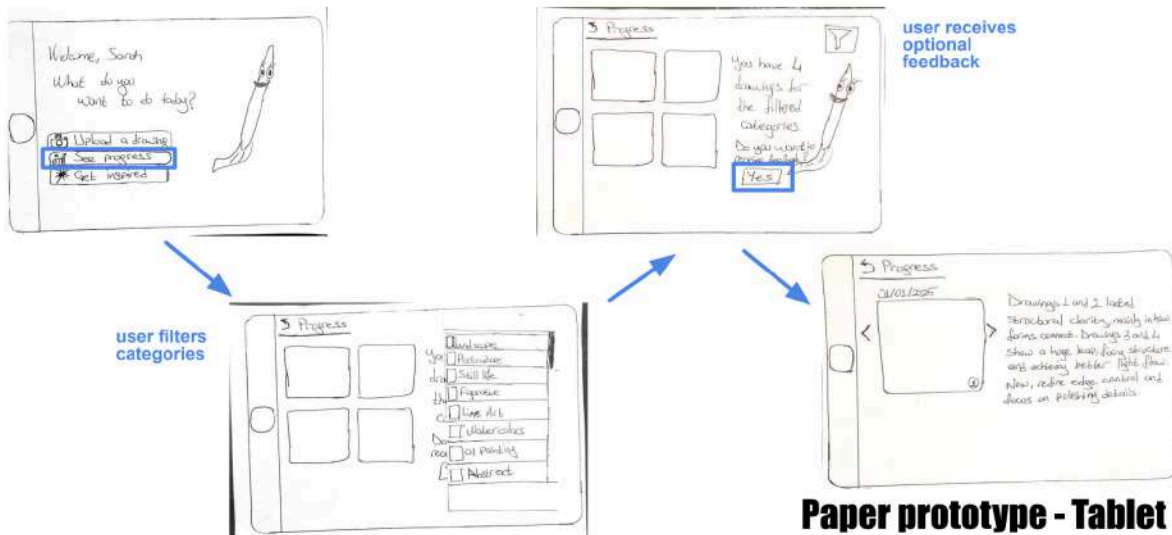
Paper prototype - Tablet

Inspiration Flow



Paper prototype - Tablet

Progress Tracking Flow



Paper prototype - Tablet

ii. Storyboard Evaluation

ii.i) Mobile Prototype

- By putting only three interaction sections to the Home Page, we aimed to attract the focus of the user to the core needs.
- By making the design of the UI simple and easy to go through, we aimed to make the usage of the app frictionless for the average user.
- We targeted the simple task with the drawing upload flow, the moderate task with the inspiration flow and the complex task with the progress tracking and feedback flow.
- Since our simple task and the base requirement is drawing upload to the system, mobile devices were a perfect modality for us since we need a camera. Since it is also very accessible it is also helpful for reaching the targeted people.

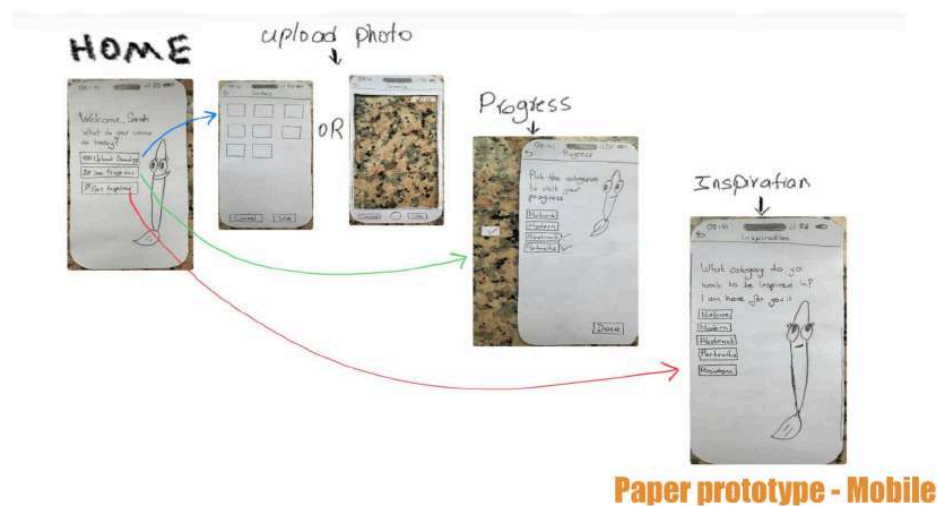
ii.ii) Tablet Prototype

- By putting only three interaction sections to the Home Page, we aimed to attract the focus of the user to the core needs.
- By making the design of the UI simple and easy to go through, we aimed to make the usage of the app frictionless for the average user.
- With the file upload flow we are handling the simple task.
- With the inspiration flow we are focusing on the moderate task
- With the progress tracking part we are targeting the complex task
- Even though it is less accessible compared to mobile devices, the tablet prototype has its own advantages. It is still satisfying our simple task's need of camera and wider screen is more helpful for our progress tracking and inspiration usages.

iii. High Level Flow

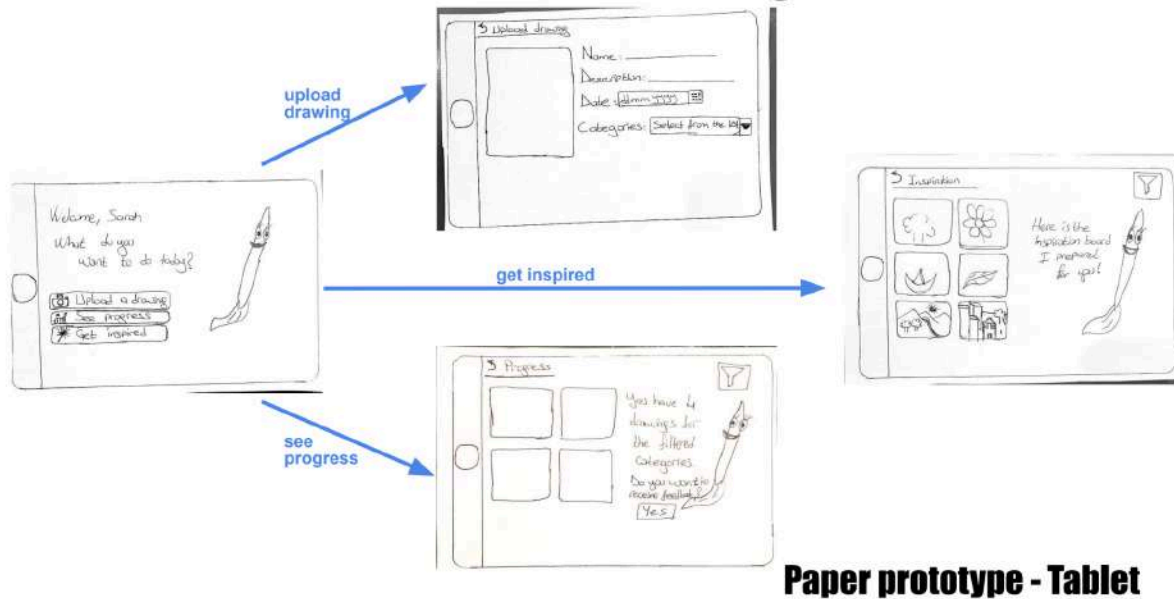
High-level flow of the Mobile Prototype

Home and Related Pages



High-level flow of the Tablet Prototype

Home and Related Pages



c. Heuristic Evaluations

i.Process

We started by taking professional feedback on our prototype and related documents. After this initial step, we iterated on each page and worked on strengthening the bond between the main tasks and the screens, while making them more intuitive. This was an important part for using the time more efficiently with the possible users who'll test the prototype.

After having our final prototype ready, in order to achieve comparable metrics, we worked on preparing a standardized evaluation speech and script for the evaluation.

ii.Execution

We received two heuristic evaluation reports on our **low-fidelity paper prototypes** (phone and tablet versions of the same core flows). The evaluators inspected the **phone flow**, and we mapped the findings to the **tablet version** wherever the same interaction exists.

How the evaluations were run

Each evaluator walked through the **same three scenarios**:

- Upload a drawing
- Find inspiration
- Check progress + get feedback

During the walkthrough, they checked the interface against **Nielsen's 10 usability heuristics** and wrote down:

- where the issue happens (screen/step),
- what goes wrong,
- which heuristic is violated (and why),
- a severity rating (minor -> major).

How we processed the results

After receiving both reports, we:

- merged duplicates,
- grouped issues by theme (upload flow, navigation, progress/feedback),
- prioritized by severity and by how often the issue appeared.

Summary of results (high level)

- Total findings across both reports: **15 raw issues**
- After merging duplicates: **13 unique violations**
- Severity distribution (using the highest severity when evaluators disagreed):

Major (3): **4**

Minor (2): **8**

Cosmetic (1): **1**

Critical (4): **0**

Main problem area: **Progress + Feedback flow** (clarity, navigation, control), followed by **Upload flow** (popup control/labels) and **efficiency** (search/metadata).

iii. Violations

Severity scale used: 1 = cosmetic, 2 = minor, 3 = major, 4 = critical

V1 - Ambiguous confirmation label

- **Heuristic:** H2 (Match between system and real world)
- **Where:** Photo upload flow (after selecting/capturing an image)
- **Violation:** Confirmation button labeled “**Use**” is unclear for a photo selection action.
- **Severity:** 2

V2 - Redundant “Back” and “Cancel”

- **Heuristic:** H8 (Aesthetic and minimalist design)
- **Where:** Upload drawing screen
- **Violation:** Having both a top-left back button and a bottom “Cancel” creates clutter and may confuse users.
- **Severity:** 1

V3 - No custom categories

- **Heuristic:** H7 (Flexibility and efficiency of use)
- **Where:** Upload screen + Progress category selection
- **Violation:** Users can only choose predefined categories; no way to create custom ones.
- **Severity:** 2

V4 - Category selection behavior unclear

- **Heuristic:** H1 (Visibility of system status)
- **Where:** Category selection (Progress / filters)
- **Violation:** Checkboxes are used, but it’s not clear if the selection is single or multi-select.
- **Severity:** 2

V5 - Required vs optional fields not indicated

- **Heuristic:** H1 (Visibility of system status)
- **Where:** Upload drawing form (name/description/category/date)
- **Violation:** The form doesn’t tell users which fields are mandatory.

- **Severity:** 2

V6 - No search in Progress (scales poorly)

- **Heuristic:** H7 (Flexibility and efficiency of use)
- **Where:** Progress drawing list
- **Violation:** Users can only filter; there's no search by name/date, which becomes a big limitation as the gallery grows.
- **Severity:** 3

V7 - Feedback prompt forces "Yes"

- **Heuristic:** H3 (User control and freedom)
- **Where:** Progress → feedback prompt dialog
- **Violation:** The dialog only offers **"Yes"** (no "No", "Later", or close), which removes user control.
- **Severity:** 3

V8 - Thumbnails lack useful metadata

- **Heuristic:** H6 (Recognition rather than recall)
- **Where:** Progress thumbnail grid
- **Violation:** Thumbnails show mainly image/date; missing name/category makes it harder to recognize items without opening them.
- **Severity:** 2

V9 - Back navigation labeling/behavior is confusing

- **Heuristics:** H3 (User control and freedom), H4 (Consistency and standards)
- **Where:** App navigation, especially in Progress flow
- **Violation:** Back behavior and labels are unclear (e.g., back label reflects current page or users can't predict where "Back" goes).
- **Severity:** 3

V10 - Carousel opens from the first item, not the selected one

- **Heuristic:** H7 (Flexibility and efficiency of use)
- **Where:** Progress → detail view carousel

- **Violation:** When a user taps a specific drawing, the detail view starts from the first drawing, forcing extra swipes.
- **Severity:** 2

V11 - Upload popup has no "Cancel/Close"

- **Heuristic:** H3 (User control and freedom)
- **Where:** Upload flow popup ("Take a photo / Select from Gallery")
- **Violation:** No clear exit (Cancel/Close). If opened by mistake, users feel trapped.
- **Severity:** 3

V12 - Feedback refers to multiple drawings but shows one at a time

- **Heuristic:** H1 (Visibility of system status)
- **Where:** Progress → detailed feedback screen
- **Violation:** Text references "Drawing 1 and 2" while only one drawing is visible, making it unclear what feedback refers to.
- **Severity:** 2

V13 - Camera screen missing standard controls

- **Heuristic:** H7 (Flexibility and efficiency of use)
- **Where:** Camera interface
- **Violation:** Lacks expected controls (flash, switch camera, gallery shortcut), reducing efficiency and familiarity.
- **Severity:** 2

d. Selection

We selected the **mobile paper prototype** as our main direction because it best matches how beginner and hobbyist artists would realistically use the app: it's always with them and gives direct access to the camera and photo gallery for **uploading drawings**. This makes the core flow (capture - save - reflect) fast and convenient, which is important for building a consistent habit.

We kept the **tablet prototype** as a secondary reference, mainly for features that benefit from a larger screen (browsing inspiration and reviewing progress), but the mobile version remained our primary baseline for design decisions.

6. Medium to High-Fidelity Prototype



a. Tools Used

We used Figma.

b. Screens Selected and Violations Fixed

The reason we chose the landing page as the first screen is that, it gives a clear image of what the app supposed to do. The user can clearly navigate starting from that, and it's important that it's intuitive.

The second page, the progress page, has been chosen because of the main functionality that the app provides, getting feedback. Here the user clearly can see the main functionalities related to the progress part.

We selected the phone prototype as the base for the medium-fidelity version for the following reasons:

- It received more detailed and actionable heuristic feedback, especially regarding navigation, user control, and efficiency.

- A mobile format is more aligned with the usage context of our project, since users are expected to upload drawings, take photos, and revisit progress frequently in everyday environments.

At this stage, no features are moved from the other prototype, as all core interactions are already present in the phone version. Instead, we focus on refining control, clarity, and flexibility based on the violations.

In the second screen we selected, we had 2 violations:

- H3 - The forced feedback popup will be redesigned to include clear options such as “Yes,” “No,” and “Maybe later”, so users are never pushed into an action unwillingly
- H7 - A search bar will be added to the Progress section to allow users to search drawings by name, date, or category.

We fixed both with adding a “No” button for the first one and the second one with adding a search bar and filter option.

c. Figma Link

[Figma Design Link](#)

d. Violation Handling Plan

For the high-fidelity prototype, we plan to resolve all severity 3 violations and as many severity 2 violations as possible, as required by the assignment. The fixes will directly build on the issues identified in the heuristic evaluations.

Fixes for Severity 3 Violations (High Priority)

H3 - User Control and Freedom

- The forced feedback popup will be redesigned to include clear options such as “Yes,” “No,” and “Maybe later”, so users are never pushed into an action unwillingly
- Back navigation will be standardized across the application. The back button will always lead to the previous logical screen, not reset the navigation flow.
- The photo source popup (camera / gallery) will include a visible Cancel or Close (X) option so users are never trapped inside a modal.

H7 - Flexibility and Efficiency of Use

- A search bar will be added to the Progress section to allow users to search drawings by name, date, or category.

Fixes for Severity 2 Violations (Medium Priority)

H1 - Visibility of System Status

- Required input fields in the upload screen will be clearly marked using placeholders or visual indicators.

- Feedback text in the Progress section will dynamically update to match the currently visible drawing.

H2 - Match Between System and the Real World

- The ambiguous “Use” button in the photo confirmation flow will be renamed to “Confirm,” “Select,” or “Upload” to reflect common mobile design standards.

H6 - Recognition Rather Than Recall

- Drawing thumbnails in the Progress grid will show at least the drawing name (and optionally the category) to allow easy recognition without opening each item.

H7 - Flexibility and Efficiency

- Users will be able to create and manage custom categories instead of being restricted to predefined ones.
- The drawing detail view will open directly to the selected drawing, instead of always starting from the first image in the carousel.

Fixes for Severity 1 Violations (Low Priority)

H8 - Aesthetic and Minimalist Design

- Redundant navigation elements such as having both Back and Cancel on the same screen will be reduced to a single, consistent control.

Excluded Violations

- The following violations were excluded from the hi-fi implementation plan, as they were based on misunderstandings or intentional design decisions:

Excluded from H1

- Unclear selection logic in category selection (checkbox vs single choice)
 - Reason: The design already uses checkboxes to explicitly represent multi-selection. No ambiguity exists.

Excluded from H4

- Back button shows current page name instead of destination page
 - Reason: The evaluator misinterpreted the page title as the back button label. The back button is icon-only. And we will already fix the back navigation issue on H3.

Excluded from H7

- Camera interface lacks standard options (flash, switch camera, gallery shortcut)
 - Reason: The app intentionally delegates image capture to the native phone camera application for simplicity and reliability.

7. High-fidelity Prototype

a. Tech Stack

- Flutter (Dart): it lets us ship a clean mobile UI quickly across devices
- Local Sqlite: It is easy to set up.
- Indie Flower font from Google Fonts supports the hand-drawn, friendly tone of the app

b. Github Repo

Mody AI Github Repository: [MODY AI](#)

Drawcompanion Github Repository: [DrawCompanion](#)

c. Design Choices

We also made a set of improvements that were not direct fixes to heuristic violations, but were done to make the prototype feel closer to a real product and to better match the emotional tone of the app.

In particular, we refined the **UI look and feel** by adding a soft background image, introducing a more consistent color palette, and smoothing the interface with rounded corners and cleaner spacing. We also changed the typography to a more **sketchy, hand-drawn style** to reflect the creative nature of the app and make it feel less “technical” and more playful. Overall, these changes were meant to create a **relaxed and welcoming vibe**.

d. Most Significant Screens

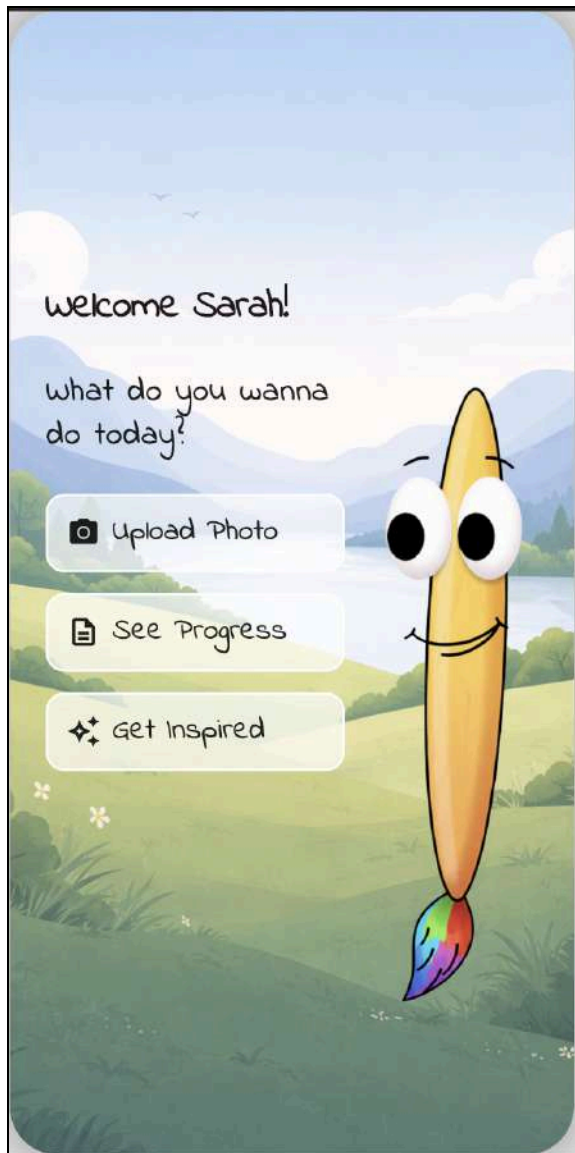
The following screens are the most significant screens. The first page is the landing page, which is important because it is the first screen the user will interact with. We tried to make it simplistic and easy to navigate. The second one is the drawing upload screen. This screen is important since it handles the base requirement of our application: uploading a drawing. We tried to make it as simple as possible. In the evaluation we got the feedback about the confusion on the category selection part. It wasn't apparently multiple choice, so in this screen we solved this issue by clarifying it better with a more widely used multiple choice tag/category selection boxes.

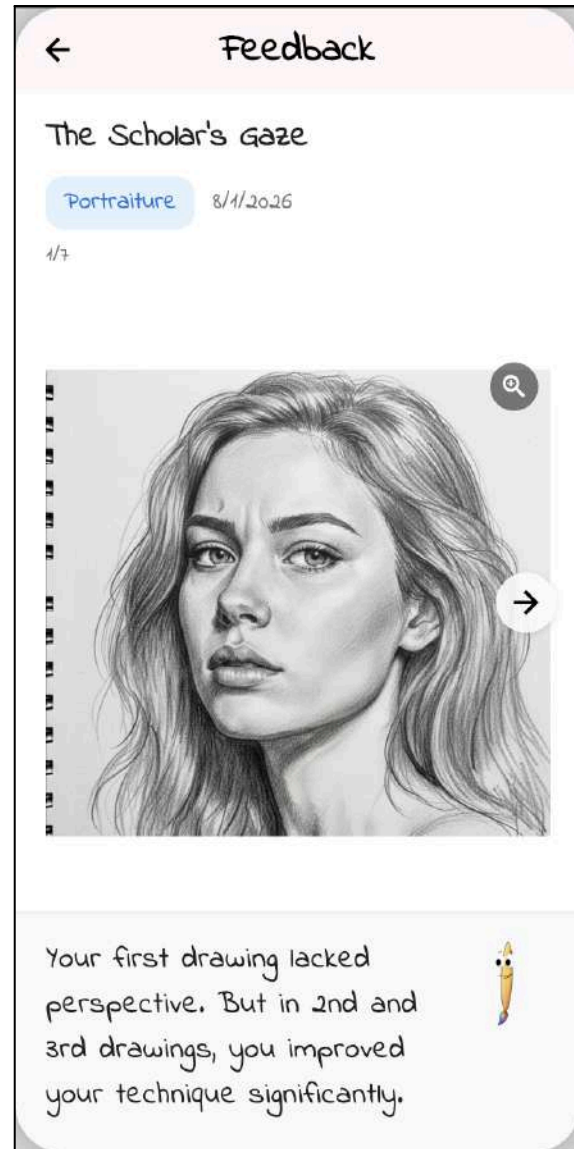
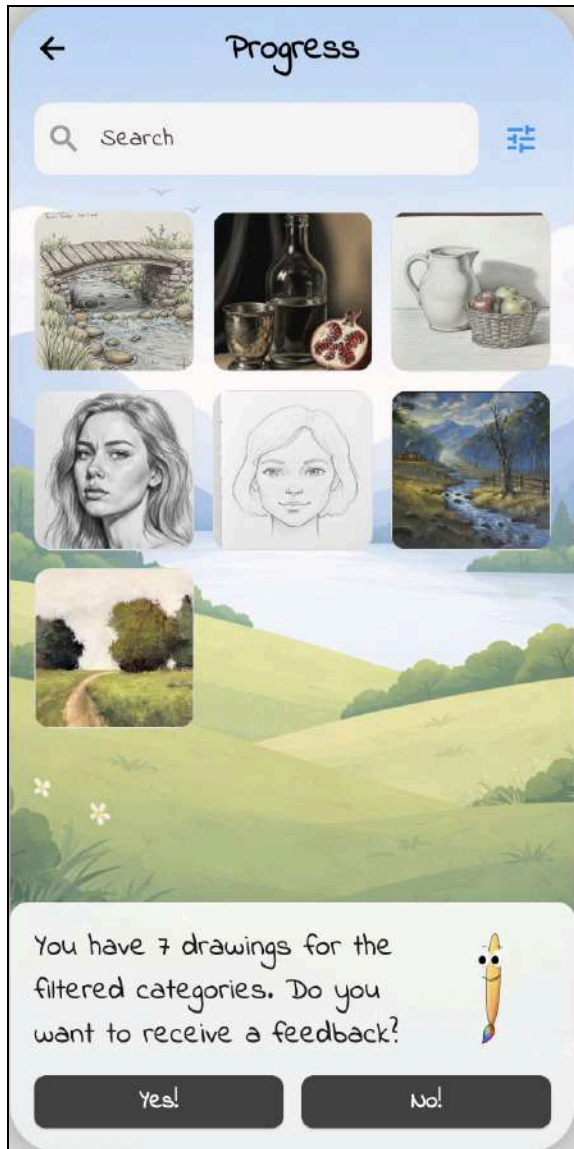
The third screen is the main page the user lands when clicked on the “See Progress” button from the landing page. This page is also significant since it is where the user checks his/her progress and we received 2 heuristic violations for this screen. The first one was a request for search and filter capability, so we added those functionalities. The second one was the “feedback pop-up” having only yes option is giving the user the feeling that she/he is not allowed to decline. To solve this second violation we added another button to the pop-up.

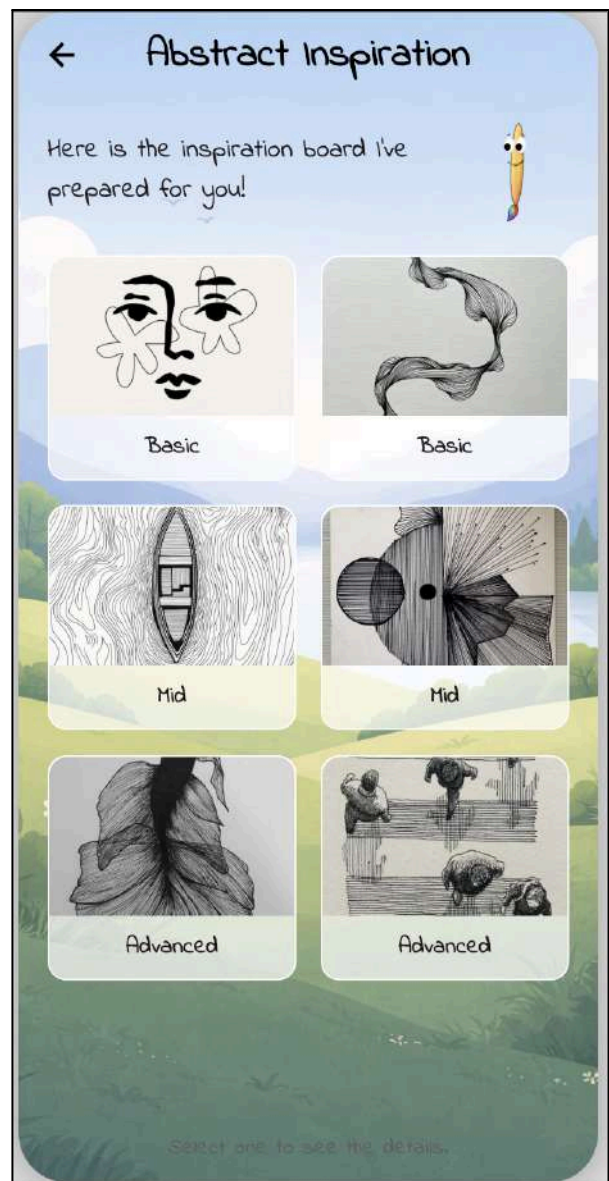
The 4th screen is the feedback screen. It is significant because it satisfies our complex task. On this screen initially we had the same design but the feedback text was related to the overall drawings of the filtered options. However we had a heuristic violation that wanted the texts to be actually related to the open drawing. We updated our design while moving to the high-fidelity prototype.

The 5th and 6th screens are the inspiration page screens. These screens are concluding our moderate task, so they are significant as well. On these screens we didn't have any heuristic violation.

Overall, again based on a heuristic evaluation, we focused on keeping consistency between the screens and made sure there is always a back button to not make the user feel stuck at any screen.







e. Standards & Constraints

We designed the prototype to feel natural on a smartphone, following common mobile app patterns and respecting the limits of small touch screens.

Our thought process included:

- Keeping the main actions (**Upload, Inspiration, Progress**) easy to reach in a few clicks.
- Using large, clear buttons and simple screens so it's easy to tap without mistakes.
- Kept text and options short and readable, since phones are often used in quick sessions.

Visual design we adopted:

- A **friendly, non-technical look** so beginners don't feel intimidated.
- **Sketchy/hand-drawn font** to match the drawing vibe.
- Added **soft colors, rounded corners, and cleaner spacing** to make the app feel calm and welcoming.
- Included a light **background illustration** to give the app a relaxed vibe.

f. Limitations

- No backend: all data is local. There is no login, or multi-device access.
- Seed data and pre-stored assets: inspiration images and sample drawings are bundled for testing, not user-generated.
- Static feedback content: feedback is currently hard-coded, not AI-generated.
- Category list is fixed: users cannot create custom categories.

8. Usability testing

a. Preparation and run

Refined Tasks

Task	Success Criteria	Methodology
Using the application upload a drawing of yours with using any method of your choice among taking a picture and uploading from camera roll.	Participants successfully uploads a photo with providing the mandatory fields and optional fields.	Think-Aloud
Find an inspiration to use in your next drawing in Nature category based on your experience level.	Participant successfully navigates through the inspiration flow and finds a drawing of her/his choice to use as inspiration.	Think-Aloud
Revisit older drawings of yours that you uploaded to check your progress and get feedback.	Participant successfully navigates to the Progress section and checks his/her older drawings and gets feedback about the progress generated by AI.	Think-Aloud

We selected Think-Aloud for the methodology because it is simple to execute and helps us to understand what the user/participant is thinking while interacting with our application through the tasks.

Quantitative metrics (per task)

- Time on task (seconds): Time needed to finish the task
- Task success (4 levels): Each task outcome is classified as:
 - 3 - Complete success: finishes the task correctly, independently, no major problems.
 - 2 - Success with minor issue: finishes correctly, but with small mistakes or inefficiency (recovers without direct help)
 - 1 - Success with major issue: finishes only with help, or finishes but with an important mistake (wrong outcome)

0 - Failure: cannot finish or gives up.

- Number of errors

Major error: prevents completing the task correctly, or requires help to finish.

Minor error: participant recovers and completes, but with confusion or extra steps.

- Number of wrong clicks (mis-clicks): Count taps/clicks that do not help progress toward the task goal (for example opening the wrong section, pressing the wrong button).

B) Subjective + qualitative metrics

- SEQ (after each task): Overall, how easy or difficult was this task to complete? Scale: 1 (Very difficult) - 7 (Very easy)
- SUS (end of test): Usability questionnaire
- Likes / dislikes / recommendations (end): Short open questions about what was clear and what was not
- Background profile (pre-test): Drawing level, drawing frequency, and whether they use drawing apps.

i. Evaluation Details

Interview 1:

- Participant: Berke Tan Muhtar -hobbyist drawer (immediate user)
- Team Members: Didem as facilitator, Yusa as observer.
- Where: A cafe in San Salvador
- How: On desktop

Interview 2:

- Participant: Gulsah Ay - hobbyist drawer (immediate user)
- Team Members: Yusa as facilitator, Didem as observer.
- Where: Home
- How: On Desktop

Interview 3:

- Participant: Sina Sohrabian (immediate user)
- Team Members: Mohammadreza as facilitator, Özgün as observer.
- Where: Starbucks in centro
- How: On mobile phone

Interview 4:

- Participant: Giada Martano -hobbyist drawer (immediate user)
- Team Members: Özgün as facilitator, Mohammadreza as observer.

- Where: A study cafe in Centro
- How: On desktop using virtual android phone

ii. Appendix

Consent Forms for Needfinding

P1 - Consent Form (PDF)

P2 - Consent Form (PDF)

P3 - Consent Form (PDF)

P4 - Consent Form (PDF)

Consent Forms for Usability Testing

- Blank Consent Form: [Consent Form Blank.pdf](#)
- P1 - [Consent Form](#)
- P2 - [Consent Form](#)
- P3 - [Consent Form](#)
- P4 - [Consent Form](#)

Observations

- P1 - [Berke Tan Muhtar - Usability Test Report](#)
- P2 - [Gulsah Ay - Usability Test Report](#)
- P3 - [Sina Sohrabian - Usability Test Report](#)
- P4 - [Giada Martano - Usability Test Report](#)

Testing Protocol

- [Test Protocol](#)

Questionnaires

- [SUS Form](#)
- [SEQ Form](#)

b. Results

i. Results

Summary

Overall, while every participant finished the Task 2 without facing any major or minor errors and almost everyone finished the Task 1 without any errors. On the other hand, on Task 3, every participant had either minor or major issues. It clearly shows our missing points and errors we have in our prototype.

Task 1:

	Time on Task	Success level	SEQ	Major Errors	Minor Errors	Wrong Clicks
Participant 1	60s	Complete success	7	0	0	0
Participant 2	50s	Success with Minor Issue	7	0	1	0
Participant 3	80s	Complete success	7	0	0	0
Participant 4	85s	Complete success	7	0	0	0

Task 2:

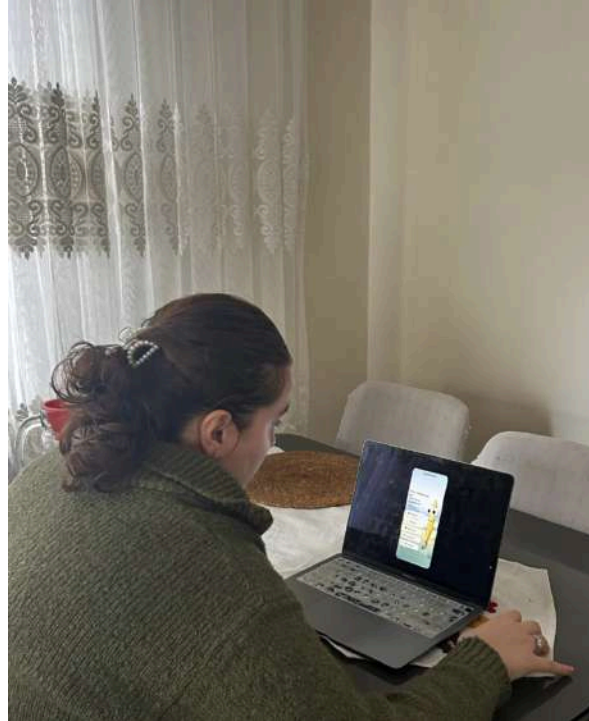
	Time on Task	Success level	SEQ	Major Errors	Minor Errors	Wrong Clicks
Participant 1	50-60s	Complete success	7	0	0	0
Participant 2	40-50s	Complete success	7	0	0	0
Participant 3	70s	Complete success	7	0	0	0
Participant 4	38s	Complete success	7	0	0	0

Task 3:

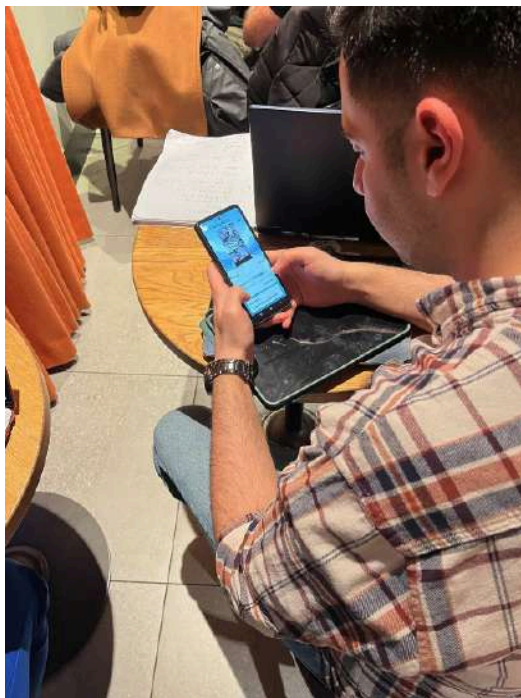
	Time on Task	Success level	SEQ	Major Errors	Minor Errors	Wrong Clicks
Participant 1	120s	Success with minor issue	6	0	1	1
Participant 2	80-90s	Success with minor issue	7	1	1	0
Participant 3	180s	Success with major issue	7	0	1	0
Participant 4	60s	Success with minor issue	6	0	1	0



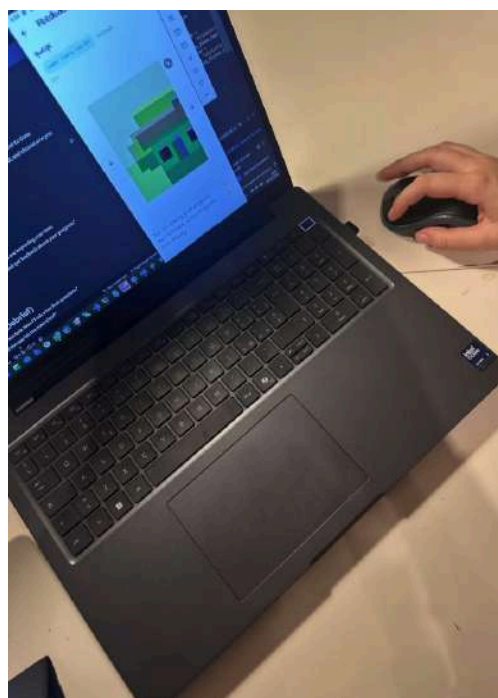
P1 doing task 1



P2 doing task 1



P3 doing task1



P4 doing task 1

Individual Observation Reports

- P1 - [Berke Tan Muhtar - Usability Test Report](#)
- P2 - [Gulsah Ay - Usability Test Report](#)
- P3 - [Sina Sohrabian - Usability Test Report](#)
- P4 - [Giada Martano - Usability Test Report](#)

i.ii Debrief Results

Participants consistently said the app felt simple, clear, and easy to navigate. The parts they found most straightforward were uploading drawings (especially the category-based organization) and the inspiration section (clear categories and levels). Most people didn't report major confusion during the session.

The main pain point was the Progress + Feedback experience. Even when users liked the idea of getting AI feedback and found it motivating, they wanted it to be clearer and more useful. Specific complaints included the unexpected behavior of the "No" button (redirecting instead of closing), and feedback that felt too generic or not technical enough. A couple of participants also mentioned that the visual tone sometimes felt a bit childish, especially on the first screen.

Overall, the app is already working well as a friendly drawing tracker and inspiration tool, but to feel complete it needs a stronger Progress/Feedback flow: clearer controls, better feedback quality, and a more consistent "grown-up" visual vibe.

ii. Discoveries

From four moderated usability tests, we learned that the prototype is **easy to understand and navigate**, especially for the first two tasks, but the **Progress + Feedback** feature needs the most improvement.

What worked well

- Overall usability was strong: participants described the app as friendly, simple, and not technical. They generally knew what to do next.
- Task 1 (Upload) was consistently successful: everyone completed it easily, and users liked the idea of organizing drawings with categories.
- Task 2 (Inspiration) was also smooth: everyone completed it quickly with high SEQ scores. Feedback here was mostly about wanting more variety/content, plus features like saving inspirations or shuffling suggestions.

What needs improvement

- Task 3 (Progress + Feedback) caused most friction: multiple participants hesitated or mis-clicked because it wasn't always clear **what the feedback referred to**, how to move through it, or when it finished.
- Feedback quality felt too generic for some users: they liked the idea of AI feedback, but expected it to be more specific and actionable (more technical tips, clearer suggestions).
- Small UI details mattered: unclear multi-category selection and confusing Yes/No popup behavior showed that minor interaction choices can affect confidence.

What is Fixed

We only encountered one major issue during the usability testings. It was that a user was unable to search when multiple categories are selected. That bug was fixed after the usability testings were completed because it affected the usability of the application.

Key takeaway

The app already works well as a **drawing journal + motivation tool**, but to feel complete it needs a clearer, more helpful **progress/feedback experience**, and a visual tone that fits both beginners and adult hobbyists.

iii. Potential Changes

1) Fix feedback navigation: clearer state, and a visible close button

Redesign the feedback flow so it behaves like a standard, predictable component:

- a progress indicator ("2 of 5 feedbacks"),
- a clear "Finish / Close" action at the end.

2) Redesign the Yes/No feedback prompt (don't redirect on "No")

If the app asks whether the user wants to view feedback:

- "No" should close the popup and keep them on the same page,
- add "Maybe later" or "Not now,"

- show a short explanation of what happens if they choose “Yes” (e.g., “You’ll see feedback based on your recent drawings”).

3) Improve feedback usefulness by making it more actionable and “teacher-like”

Update feedback content to feel more helpful and concrete, for example:

- add “Try next” mini-exercises or prompts,
- allow users to choose feedback style: “Encouraging / Technical / Fun.”

4) Fix category multi-select clarity

Change (UI + logic):

- Change label “Category” to “Categorie(s)”.
- Add helper text (“You can select multiple”).

5) Add sorting and date filtering in Progress (scale the feature for real use)

In Progress:

- allow sorting (newest/oldest),
- add a date range filter (last week/month/custom),
- show a simple timeline view or month grouping.

7) Make inspiration more “usable” by adding save/shuffle and increasing variety

In the Inspiration section:

- add Save to gallery / Favorites for inspiration items,
- add a Shuffle button for mixed inspiration,
- increase the number/variety of inspirations per category (or show “load more”).

8) Adjust the visual tone to feel less “kids-only” while staying friendly

Keep the friendly vibe but make the style slightly more mature:

- refine illustration/background intensity,
- use cleaner typography hierarchy

9. Conclusions

a. Learnings

We started by changing our way of thinking about needfinding. We slowly understood to think about the core needs, rather than the functionalities directly. This gave us the opportunity to work together on a topic that we're not accustomed to. With the fact that the proposed project had to be unrelated to any of us, it was a challenge to find a problem that's not part of us.

We learned the ways people can interact with the AI "beyond communication". This lab team was definitely a good match for all of us.

About our project, as we were not direct members of the domain, we gained insights through the needfinding phase and learned to work on a domain we do not know much about in the beginning. It was challenging at first to see that what we have in mind does not really match with what people understand by our designs or ideas. We learned to lean more on conventions of human computer architecture and we learned to be more direct in terms of tasks and stories.

b. Feedbacks

At the beginning of the semester, we needed some time to get to understand how each of us work, communicate and think. This process required time, but the time we had for the first task was pretty limited. This created chaos for the first task and we needed some time to recover from that and build a robust foundation for the rest of the project.

The fact that all of us are developers, we underestimated the reporting part of the tasks, which caused unbalanced working loads within the group. This problem slowly faded away as we went through new tasks. Afterwards, we splitted all the tasks into small chunks and worked on them in parallel. This made the reporting part also easy to split and complete for all of us.